Projector-Pulay Force Corrections in DFT+eDMFT

Finite band-window response, differentiation of projector normalization, and correction of the logarithmic force term

Detailed derivation and implementation plan

Scope. This note derives the terms that are missing when the normalized DFT+eDMFT projector is differentiated with respect to an atomic displacement. It is intended to supplement the existing LAPW force derivation and to guide a future implementation in `src/dmft2`. No source-code modifications are made here.

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Contents

1 Purpose and central result **1**

2 Notation and conventions **1**

3 Why the stationary functional is differentiated **2**

3.1 Schematic Klein functional 2

3.2 Why no two-particle vertex appears 3

4 What is already present in the force derivation **3**

5 A projector that is finite at the muffin-tin boundary **4**

6 Finite fixed-index band window **4**

6.1 Projector onto the selected band subspace 4

6.2 Response of a generalized LAPW eigenproblem 5

6.3 Window contribution to the raw-projector derivative 6

6.4 Degeneracies and window-boundary crossings 6

7 Normalization of the finite-window projector **6**

7.1 Definition 6

7.2 Derivative of the overlap matrix 7

7.3 Derivative of the inverse square root 7

7.4 Derivative of the normalization identity 8

8 Complete projector force and its three parts **8**

8.1 Existing direct rigid-projector term 9

8.2 New finite-window term 9

8.3 New normalization-response term 9

8.4 Efficient local-space contraction for the X_A force 9

9 Correction of the naive logarithmic-force normalization **10**

9.1 What the code currently does 10

9.2 Why the division cannot be correct 10

10 Additional band-dependent projector normalization **11**

11 Exact and practical forms of the band response **12**

11.1 The induced Kohn-Sham potential 12
11.2 Stored-band sum versus Sternheimer equation 12

12 Mapping to the present `dmft2` implementation **12**

13 Proposed implementation sequence **13**

14 Memory and computational cost **15**

15 Verification strategy **15**

15.1 Algebraic checks at each geometry 15
15.2 Finite-difference checks of intermediate objects 16
15.3 Band-completeness test 16
15.4 Acoustic sum rule 16
15.5 Recommended sequence of physical tests 16

16 Controlled approximations and open questions **17**

17 Compact formula sheet **17**

18 References and implementation sources **18**

1 Purpose and central result

The existing DFT+eDMFT force derivation correctly starts from a stationary Klein functional rather than differentiating a nonstationary total-energy expression. It also derives the ordinary LAPW Pulay terms and the explicit derivative of a rigid projector attached to a muffin-tin sphere. The numerical evidence, however, shows two additional sensitivities:

- the sum of forces changes strongly when the matrix `SOlapm.dat` is used or replaced by the identity;
- the force sum improves as the number of bands in the DMFT projector window is increased.

Both observations point to the same object: the normalized projector defined inside a finite Kohn-Sham band subspace. If the raw projector is denoted by $U^{\mathbf{k}}$ and its normalization by $X = S^{-1/2}$, then

$$\overline{U}^{\mathbf{k}} = U^{\mathbf{k}} X, \qquad S = \sum_{\mathbf{k}} w_{\mathbf{k}} U^{\mathbf{k}\dagger} U^{\mathbf{k}}. \tag{1}$$

The exact displacement derivative is

$$\boxed{\partial_A \overline{U}^{\mathbf{k}} = U_A^{\mathbf{k},\mathrm{dir}} X + U_A^{\mathbf{k},\mathrm{win}} X + U^{\mathbf{k}} X_A, \qquad X_A \equiv \partial_A X.} \tag{2}$$

The current force channels contain essentially the first term, with X held fixed. The second term describes mixing between bands inside and outside the fixed-index DMFT window. The third term differentiates the nontrivial matrix normalization. These two missing terms can produce forces on an uncorrelated atom even when the local correlated orbital itself is centered on another atom.

The corresponding projector-force correction is

$$\boxed{\Delta F_A^{\mathrm{proj}} = F_A^{\mathrm{win}} + F_A^{X},} \tag{3}$$

after removing an additional, erroneous elementwise division by diagonal entries of `SOlapm` in the present logarithmic-force code.

Most important practical statement. The normalized ordinary projector and all three existing vector-projector channels are already multiplied by `SOlapm`. Therefore the force built from those channels must not later be divided by products of diagonal `SOlapm` elements. The correct post-factor is one.

2 Notation and conventions

An atomic displacement variable is abbreviated as

$$A \equiv (A, \alpha), \qquad \partial_A \equiv \frac{\partial}{\partial R_{A\alpha}}, \tag{4}$$

where the first A labels the atom and $\alpha = x, y, z$. To avoid confusion, a will be used for a Kohn-Sham band outside the DMFT window.

At a given $\mathbf{k}$:

- $n, n' \in W_{\mathbf{k}}$ are bands in the fixed-index DMFT window;
- $a, b \in R_{\mathbf{k}}$ are retained bands outside that window;
- m, m' are correlated local-orbital indices;
- $w_{\mathbf{k}}$ is the Brillouin-zone weight, with $\partial_A w_{\mathbf{k}} = 0$;
- Tr_b and $\mathrm{Tr}_{\mathrm{loc}}$ denote traces in band and local spaces.

The raw band-by-local projector is

$$U^{\mathbf{k}}_{nm} = \langle \psi_{n\mathbf{k}} | \phi_m \rangle. \tag{5}$$

The self-energy that is embedded into the lattice problem is

$$\widetilde{\Sigma}(i\omega_n) = \Sigma(i\omega_n) - V_{\mathrm{DC}}, \tag{6}$$

and the embedded band-space potential is

$$V_{\Sigma,\mathbf{k}}(i\omega_n) = \overline{U}^{\mathbf{k}} \widetilde{\Sigma}(i\omega_n) \overline{U}^{\mathbf{k}\dagger}. \tag{7}$$

All matrix equations below apply separately to each correlated block (`icix`) unless otherwise stated.

3 Why the stationary functional is differentiated

3.1 Schematic Klein functional

A convenient schematic form of the DFT+eDMFT functional is

$$\begin{aligned} \Gamma[G, \rho] = & -T\,\mathrm{Tr}\ln[-G^{-1}] - T\,\mathrm{Tr}[(G_0^{-1} - G^{-1})G] \\ & + E_{\mathrm{H}}[\rho] + E_{\mathrm{xc}}[\rho] - \mathrm{Tr}[(V_{\mathrm{H}} + V_{\mathrm{xc}})\rho] \\ & + \Phi_{\mathrm{DMFT}}[G_{\mathrm{loc}}] - \Phi_{\mathrm{DC}}[n_{\mathrm{loc}}] + E_{\mathrm{nuc}}. \end{aligned} \tag{8}$$

The precise bookkeeping used by the code can be rearranged without changing the following argument. At the self-consistent solution,

$$\frac{\delta\Gamma}{\delta G} = 0, \qquad \frac{\delta\Gamma}{\delta\rho} = 0, \tag{9}$$

including the impurity stationarity condition

$$\frac{\delta\Phi_{\mathrm{DMFT}}}{\delta G_{\mathrm{loc}}} = \Sigma. \tag{10}$$

Consequently, for a nuclear coordinate,

$$\frac{\mathrm{d}\Gamma_*}{\mathrm{d}R_{A\alpha}} = \left.\frac{\partial\Gamma}{\partial R_{A\alpha}}\right|_{G,\rho}, \tag{11}$$

where the star denotes the stationary solution. The implicit derivatives $\mathrm{d}G/\mathrm{d}R_{A\alpha}$ and $\mathrm{d}\Sigma/\mathrm{d}R_{A\alpha}$ cancel between stationary terms. This is the envelope theorem in the present functional setting.

3.2 Why no two-particle vertex appears

If one differentiated the Dyson equation directly, one would generate $\partial_A\Sigma = (\delta\Sigma/\delta G_{\mathrm{loc}})\partial_A G_{\mathrm{loc}}$ and hence the impurity two-particle vertex. In the stationary-functional derivative this response is multiplied by the functional stationarity condition and cancels. Only explicit dependence on the nuclear coordinate remains. In particular, the explicit projector dependence in Eq. (7) survives.

Thus, in the projector term, G and $\widetilde{\Sigma}$ are held fixed while the projector is differentiated:

$$\partial_A V_{\Sigma,\mathbf{k}}\big|_{G,\widetilde{\Sigma}} = D_A^{\mathbf{k}}\widetilde{\Sigma}\overline{U}^{\mathbf{k}\dagger} + \overline{U}^{\mathbf{k}}\widetilde{\Sigma}D_A^{\mathbf{k}\dagger}, \tag{12}$$

where

$$D_A^{\mathbf{k}} \equiv \partial_A\overline{U}^{\mathbf{k}}. \tag{13}$$

With the sign convention used in the existing force notes, the force is minus the derivative of the stationary functional:

$$\boxed{F_A^{\mathrm{proj}} = -T\sum_{\mathbf{k},i\omega_n} w_{\mathbf{k}} e^{i\omega_n 0^+}\,\mathrm{Tr}_b\Big[G_{\mathbf{k}}(i\omega_n)\left(D_A^{\mathbf{k}}\widetilde{\Sigma}(i\omega_n)\overline{U}^{\mathbf{k}\dagger} + \overline{U}^{\mathbf{k}}\widetilde{\Sigma}(i\omega_n)D_A^{\mathbf{k}\dagger}\right)\Big].} \tag{14}$$

It is safest to retain both terms. At a single positive Matsubara frequency they need not be literal complex conjugates, although the complete frequency sum is real.

4 What is already present in the force derivation

The existing derivation writes the variation of the stationary functional as

$$\begin{aligned}\delta\Gamma = {} & \mathrm{Tr}\Big[\rho^{\mathrm{DMFT}}A^{0\dagger}(\delta H^0)A^0\Big] - \mathrm{Tr}\Big[(\rho\epsilon)^{\mathrm{DMFT}}A^{0\dagger}(\delta O)A^0\Big] \\ & - \mathrm{Tr}(\delta V_{\mathrm{KS}}\rho) - \sum_A F_A^{\mathrm{HF}}\delta R_A\delta\Gamma_{\mathrm{proj}}.\end{aligned} \tag{15}$$

For the LAPW Hamiltonian and overlap, it obtains

$$\begin{aligned}H_A^0 = {} & R_A^{(1)} - R_A^{(2)} + R_A^{(3)} + R_A^{(4)} - \int_{\mathrm{MT}}\mathrm{d}^3r\, V_{\mathrm{KS}}(\mathbf{r})\nabla_A[\chi^*_{\mathbf{K}'}(\mathbf{r})\chi_{\mathbf{K}}(\mathbf{r})] \\ & + \langle\chi_{\mathbf{K}'}|V_{\mathrm{KS},A}|\chi_{\mathbf{K}}\rangle,\end{aligned} \tag{16}$$

$$O_A = Q_A^{(1)} - Q_A^{(2)}. \tag{17}$$

These give the ordinary Hamiltonian/overlap Pulay force. The same *off-diagonal* matrices will also be reused below to calculate the response of the finite band window. This reuse must not be confused with adding the ordinary Pulay force twice.

For the rigid projector product the force notes show

$$\partial_A \left[\langle\chi_{\mathbf{K}'}|\phi_{m'}\rangle\langle\phi_m|\chi_{\mathbf{K}}\rangle\right] = iu(K_\alpha - K'_\alpha)\langle\chi_{\mathbf{K}'}|\phi_{m'}\rangle\langle\phi_m|\chi_{\mathbf{K}}\rangle. \tag{18}$$

The vector projectors stored in channels 2–4 are an efficient representation of this direct rigid-translation derivative. Equation (18) does not, however, differentiate either the selected Kohn-Sham subspace or its matrix normalization.

5 A projector that is finite at the muffin-tin boundary

Let a local projector centered at $\mathbf{R}_A$ be represented schematically by

$$\phi_A(\mathbf{r}) = f(\mathbf{r} - \mathbf{R}_A)\Theta(R_{\mathrm{MT}} - |\mathbf{r} - \mathbf{R}_A|), \tag{19}$$

where f need not vanish at R_{MT}. Its distributional derivative is

$$\partial_A\phi_A = -\nabla_\alpha f\,\Theta(R_{\mathrm{MT}} - r_A) + f\,\hat{n}_\alpha\delta(r_A - R_{\mathrm{MT}}). \tag{20}$$

The second term is a surface delta function. When the derivative of the LAPW basis and the integration-by-parts surface term are treated consistently, this surface contribution cancels the corresponding boundary term. The final rigid translation result is Eq. (18); there is no additional, independent "projector surface force" at fixed R_{MT}.

The nonzero boundary value is nevertheless numerically important: it makes the projector less smooth and therefore increases the number of Kohn-Sham bands needed to represent its response accurately. This is consistent with the observed sensitivity to the band window.

The surface cancellation assumes a rigidly translated sphere with fixed radius. A derivative with respect to R_{MT}, or an inconsistent truncation of one of the two surface contributions, is a different problem.

6 Finite fixed-index band window

6.1 Projector onto the selected band subspace

The selected subspace is

$$Q_{\mathbf{k}} = \sum_{n\in W_{\mathbf{k}}} |\psi_{n\mathbf{k}}\rangle\langle\psi_{n\mathbf{k}}|. \tag{21}$$

For a fixed-index window such as $W_{\mathbf{k}} = \{65, \ldots, 164\}$, no derivative of a band-selection step function is present. The indices remain fixed. What does change is the subspace spanned by those eigenvectors:

$$Q_{A,\mathbf{k}} = \sum_{n \in W_{\mathbf{k}}} \left(|\psi_{n\mathbf{k},A}\rangle\langle\psi_{n\mathbf{k}}| + |\psi_{n\mathbf{k}}\rangle\langle\psi_{n\mathbf{k},A}| \right). \tag{22}$$

Choose a parallel-transport gauge inside $W_{\mathbf{k}}$, or simply observe that unitary rotations entirely within $W_{\mathbf{k}}$ cancel from $Q_{A,\mathbf{k}}$. Only mixing across the window boundary remains:

$$\boxed{Q_{A,\mathbf{k}} = \sum_{\substack{n \in W_{\mathbf{k}} \\ a \notin W_{\mathbf{k}}}} \left[|\psi_{a\mathbf{k}}\rangle C^{A,\mathbf{k}}_{an} \langle\psi_{n\mathbf{k}}| + |\psi_{n\mathbf{k}}\rangle C^{A,\mathbf{k}*}_{an} \langle\psi_{a\mathbf{k}}| \right].} \tag{23}$$

In practice a is first restricted to the stored complement $R_{\mathbf{k}}$ in `case.vector` or `case.vectorso`. Convergence is tested by requesting more LAPW1/LAPWSO bands.

6.2 Response of a generalized LAPW eigenproblem

The coefficient vectors satisfy

$$H_{\mathbf{k}} A_{n\mathbf{k}} = \epsilon_{n\mathbf{k}} O_{\mathbf{k}} A_{n\mathbf{k}}, \qquad A^{\dagger}_{a\mathbf{k}} O_{\mathbf{k}} A_{n\mathbf{k}} = \delta_{an}. \tag{24}$$

Differentiate the eigenvalue equation and project from the left with $A^{\dagger}_{a\mathbf{k}}$, for $a \neq n$:

$$\begin{aligned} 0 = {} & A^{\dagger}_{a\mathbf{k}} (H_{A,\mathbf{k}} - \epsilon_{n\mathbf{k}} O_{A,\mathbf{k}}) A_{n\mathbf{k}} \\ & + (\epsilon_{a\mathbf{k}} - \epsilon_{n\mathbf{k}}) A^{\dagger}_{a\mathbf{k}} O_{\mathbf{k}} A_{n\mathbf{k},A}. \end{aligned} \tag{25}$$

Therefore the covariant interband mixing coefficient is

$$\boxed{C^{A,\mathbf{k}}_{an} \equiv A^{\dagger}_{a\mathbf{k}} O_{\mathbf{k}} A_{n\mathbf{k},A} = \frac{A^{\dagger}_{a\mathbf{k}} (H_{A,\mathbf{k}} - \epsilon_{n\mathbf{k}} O_{A,\mathbf{k}}) A_{n\mathbf{k}}}{\epsilon_{n\mathbf{k}} - \epsilon_{a\mathbf{k}}}.} \tag{26}$$

Define the rectangular matrices

$$H^{R,W}_A = A^{\dagger}_R H_A A_W, \tag{27}$$
$$O^{R,W}_A = A^{\dagger}_R O_A A_W, \tag{28}$$
$$M^{R,W}_A = H^{R,W}_A - O^{R,W}_A \,\mathrm{diag}(\epsilon_W). \tag{29}$$

Then

$$C^{A}_{an} = \frac{M^{A}_{an}}{\epsilon_n - \epsilon_a}, \qquad a \in R, \quad n \in W. \tag{30}$$

This is an explicit sum over stored excluded bands, not a Sternheimer solve. The cost is modest because the vector files already contain many more bands than the DMFT window.

6.3 Window contribution to the raw-projector derivative

From Eq. (5),

$$\partial_A U^{\mathbf{k}}_{nm} = \langle \psi_{n\mathbf{k},A} | \phi_m \rangle + \langle \psi_{n\mathbf{k}} | \phi_{m,A} \rangle. \tag{31}$$

The first term, restricted to excluded-band mixing, is

$$\boxed{(U^{\mathbf{k},\mathrm{win}}_A)_{nm} = \sum_{a \in R_{\mathbf{k}}} C^{A,\mathbf{k}*}_{an} U^{\mathbf{k}}_{am}.} \tag{32}$$

The second term, together with the LAPW moving-basis convention used to represent it, is denoted $U^{\mathbf{k},\mathrm{dir}}_A$. Thus

$$U^{\mathbf{k}}_A = U^{\mathbf{k},\mathrm{dir}}_A + U^{\mathbf{k},\mathrm{win}}_A. \tag{33}$$

For an uncorrelated displaced atom, the physical local-orbital derivative is usually zero,

$$\phi_{m,A} = 0, \tag{34}$$

but $U^{\mathbf{k},\mathrm{win}}_A$ is generally nonzero because the Kohn-Sham subspace responds to every atomic displacement. This is the simplest mechanism by which the missing projector force appears on a noncorrelated atom.

6.4 Degeneracies and window-boundary crossings

Equation (26) becomes ill-conditioned if an included and an excluded band are nearly degenerate. This is not a failure of perturbation theory inside a well-defined isolated subspace; it signals that a fixed-index boundary cuts a nearly degenerate manifold. Robust options are:

- enlarge the DMFT window so the entire manifold is included;
- treat the nearly degenerate included/excluded group by a block resolvent rather than individual denominators;
- monitor the smallest $|\epsilon_n - \epsilon_a|$ at the boundary and warn when it falls below a chosen tolerance.

For the initial implementation, enlarging the window is the simplest and most transparent policy.

7 Normalization of the finite-window projector

7.1 Definition

The raw-projector overlap is

$$\boxed{S_{mm'} = \sum_{\mathbf{k}} w_{\mathbf{k}} \sum_{n \in W_{\mathbf{k}}} U^{\mathbf{k}*}_{nm} U^{\mathbf{k}}_{nm'} = \sum_{\mathbf{k}} w_{\mathbf{k}} \langle \phi_m | Q_{\mathbf{k}} | \phi_{m'} \rangle.} \tag{35}$$

The matrix used in the code is

$$X = S^{-1/2}, \tag{36}$$

and the normalized projector is

$$\overline{U}^{\mathbf{k}} = U^{\mathbf{k}} X. \tag{37}$$

It obeys

$$\boxed{\sum_{\mathbf{k}} w_{\mathbf{k}} \overline{U}^{\mathbf{k}\dagger} \overline{U}^{\mathbf{k}} = I.} \tag{38}$$

7.2 Derivative of the overlap matrix

Since the $\mathbf{k}$ weights are constant,

$$\boxed{S_A = \sum_{\mathbf{k}} w_{\mathbf{k}} \left(U_A^{\mathbf{k}\dagger} U^{\mathbf{k}} + U^{\mathbf{k}\dagger} U_A^{\mathbf{k}} \right).} \tag{39}$$

In components,

$$(S_A)_{mm'} = \sum_{\mathbf{k}} w_{\mathbf{k}} \sum_{n \in W_{\mathbf{k}}} [(U_{nm}^A)^* U_{nm'} + U_{nm}^* U_{nm'}^A]. \tag{40}$$

Using Eq. (33),

$$S_A = S_A^{\mathrm{dir}} + S_A^{\mathrm{win}}. \tag{41}$$

Even when the displaced atom is not correlated, S_A^{win} is in general nonzero.

7.3 Derivative of the inverse square root

Let

$$Y = S^{1/2} = X^{-1}. \tag{42}$$

Differentiate $XSX = I$:

$$X_A Y + X S_A X + Y X_A = 0. \tag{43}$$

Therefore X_A is the solution of the Hermitian Sylvester equation

$$\boxed{Y X_A + X_A Y = -X S_A X.} \tag{44}$$

This is computationally inexpensive because the matrices live only in the correlated-orbital space.

An equivalent explicit formula follows from

$$S = V\,\mathrm{diag}(s_1, \ldots, s_d)V^\dagger. \tag{45}$$

Let

$$\widehat{S}_A = V^\dagger S_A V. \tag{46}$$

Then

$$\boxed{X_A = V\left(F \circ \widehat{S}_A\right)V^\dagger,} \tag{47}$$

where $\circ$ is the elementwise product and

$$F_{aa} = -\frac{1}{2s_a^{3/2}}, \tag{48}$$

$$F_{ab} = -\frac{1}{\sqrt{s_a}\sqrt{s_b}(\sqrt{s_a}+\sqrt{s_b})}, \qquad a \neq b. \tag{49}$$

Equation (47) is the Frechet derivative of $S^{-1/2}$; it remains well behaved when s_a and s_b approach each other because the off-diagonal divided difference tends smoothly to the diagonal expression.

7.4 Derivative of the normalization identity

The strongest local consistency check follows by differentiating Eq. (38):

$$\boxed{\sum_{\mathbf{k}} w_{\mathbf{k}}\left[D_A^{\mathbf{k}\dagger}\overline{U}^{\mathbf{k}} + \overline{U}^{\mathbf{k}\dagger}D_A^{\mathbf{k}}\right] = 0.} \tag{50}$$

Failure of this identity directly diagnoses an incomplete S_A, an incorrect X_A, or inconsistent $\mathbf{k}$ weights/projector conventions.

8 Complete projector force and its three parts

Insert Eq. (2) into Eq. (14). Define

$$D_A^{\mathbf{k},\mathrm{dir}} = U_A^{\mathbf{k},\mathrm{dir}}X, \tag{51}$$

$$D_A^{\mathbf{k},\mathrm{win}} = U_A^{\mathbf{k},\mathrm{win}}X, \tag{52}$$

$$D_A^{\mathbf{k},X} = U^{\mathbf{k}}X_A. \tag{53}$$

Then

$$F_A^{\mathrm{proj}} = F_A^{\mathrm{dir}} + F_A^{\mathrm{win}} + F_A^{X}. \tag{54}$$

8.1 Existing direct rigid-projector term

After the erroneous post-normalization is removed, the present vector-projector force represents

$$F_A^{\mathrm{dir}} = -T \sum_{\mathbf{k},i\omega_n} w_{\mathbf{k}} \,\mathrm{Tr}_b \Big[G_{\mathbf{k}} (U_A^{\mathbf{k},\mathrm{dir}} X \widetilde{\Sigma} X U^{\mathbf{k}\dagger} + U^{\mathbf{k}} X \widetilde{\Sigma} X U_A^{\mathbf{k},\mathrm{dir}\dagger}) \Big]. \tag{55}$$

This statement assumes that the existing vector channels fully represent the direct derivative in the same LAPW convention used by the force derivation.

8.2 New finite-window term

$$\boxed{F_A^{\mathrm{win}} = -T \sum_{\mathbf{k},i\omega_n} w_{\mathbf{k}} \,\mathrm{Tr}_b \Big[G_{\mathbf{k}} (U_A^{\mathbf{k},\mathrm{win}} X \widetilde{\Sigma} X U^{\mathbf{k}\dagger} + U^{\mathbf{k}} X \widetilde{\Sigma} X U_A^{\mathbf{k},\mathrm{win}\dagger}) \Big].} \tag{56}$$

All frequency arguments $i\omega_n$ are implicit in Eq. (56). This term must be evaluated for every displaced atom, not only correlated atoms.

8.3 New normalization-response term

$$\boxed{F_A^{X} = -T \sum_{\mathbf{k},i\omega_n} w_{\mathbf{k}} \,\mathrm{Tr}_b \Big[G_{\mathbf{k}} (U^{\mathbf{k}} X_A \widetilde{\Sigma} X U^{\mathbf{k}\dagger} + U^{\mathbf{k}} X \widetilde{\Sigma} X_A U^{\mathbf{k}\dagger}) \Big].} \tag{57}$$

Because X_A depends on the global $\mathbf{k}$ sum in S_A, it is not available until after a Brillouin-zone/MPI reduction. The next subsection shows how to avoid rereading all vector files.

8.4 Efficient local-space contraction for the X_A force

Define

$$L_{\mathbf{k}}(i\omega_n) = U^{\mathbf{k}\dagger} G_{\mathbf{k}}(i\omega_n) U^{\mathbf{k}}. \tag{58}$$

Cyclically permuting the band trace in Eq. (57) gives

$$\boxed{K = T \sum_{\mathbf{k},i\omega_n} w_{\mathbf{k}} \left[\widetilde{\Sigma}(i\omega_n) X L_{\mathbf{k}}(i\omega_n) + L_{\mathbf{k}}(i\omega_n) X \widetilde{\Sigma}(i\omega_n) \right],} \tag{59}$$

followed by the small-matrix contraction

$$\boxed{F_A^X = -\operatorname{Tr}_{\mathrm{loc}}(X_A K).} \tag{60}$$

Thus K can be accumulated during the main $\mathbf{k}$ loop. After S_A and K are reduced, one solves Eq. (44) and evaluates Eq. (60). No second vector-file pass is required for this term.

The existing code separates static and dynamic self-energy pieces and applies a high-frequency correction to Matsubara sums. The same quadrature and tail logic must be used for K and for the new window-force contraction; otherwise the new term will not be numerically consistent with the old projector force.

9 Correction of the naive logarithmic-force normalization

9.1 What the code currently does

The present sequence is:

1. `Build_DMFT_Projector` constructs the ordinary projector and, when forces are requested, three vector-projector channels.
2. `RenormalizeTrans` right-multiplies *every* channel by `SOlapm`$= X$.
3. `CompressSigmaTransformation2` constructs `lgTrans` from those already normalized channels.
4. `cmpLogGdloc` constructs the static and dynamic $\mathrm{Tr}(G\widetilde{\Sigma})$ force contributions.
5. The result is then divided by `SOlapm`(p,p)`SOlapm`(q,q).

In equations, steps 1–3 already use

$$\overline{U} = UX, \qquad D_A^{(0)} = U_A^{\mathrm{dir}} X. \tag{61}$$

The force contraction is therefore already a normalized-projector expression. The later scalar factor

$$r_{pq}^{\mathrm{old}} = \frac{1}{X_{pp} X_{qq}} \tag{62}$$

partially removes the normalization a second time.

9.2 Why the division cannot be correct

There are three independent reasons:

1. X has already been applied to both ordinary and derivative channels.

2. X is generally a full matrix. No elementwise division by $X_{pp}X_{qq}$ can invert the matrix transformation.
3. The transformation of a derivative is not an inverse rescaling but Eq. (2), which contains the additional matrix term UX_A.

The correct post-factor is

$$\boxed{r_{pq}^{\text{correct}} = 1.} \tag{63}$$

No diagonal or elementwise `SOlapm` division should be applied to the force after `lgTrans`/`Gdloc` has been built from normalized channels.

This correction concerns the projector force obtained from the logarithmic/Klein-functional term. It does not prescribe multiplying or dividing the scalar value of `logG` itself by `SOlapm`.

There are two additional implementation warnings:

- a diagnostic expression indexes `SOlapm` with a compressed self-energy index, which need not be a local-orbital index;
- `Olap_Renormalize` is currently hardwired true inside `cmpLogGdloc`, even though the allocation and use of `SOlapm` are controlled elsewhere by `Qrenormalize`.

Removing the redundant force rescaling also removes these inconsistencies from the force path.

10 Additional band-dependent projector normalization

For projector types 2 and 4, the code multiplies the raw projector by a band-dependent factor called `dff`. Schematically,

$$U_{nm} = d_n u_{nm}, \qquad d_n = \sqrt{\frac{s_{1n}}{s_{2n}}}, \tag{64}$$

where s_{1n} and s_{2n} are positive norm-like quantities. Its derivative is

$$\boxed{(U_A)_{nm} = d_n (u_A)_{nm} + d_{n,A} u_{nm},} \tag{65}$$

with

$$\boxed{d_{n,A} = \frac{d_n}{2}\left(\frac{s_{1n,A}}{s_{1n}} - \frac{s_{2n,A}}{s_{2n}}\right).} \tag{66}$$

The current derivative channels are multiplied by the same d_n as the ordinary projector, which represents the first term only. If these projector types are used, $d_{n,A}u_{nm}$ is another missing direct term. It should be included in U_A^{dir} before constructing S_A and the force.

If the numerical implementation uses absolute values in Eq. (64), the derivative should be written in terms of $\partial_A \ln|s_i|$. Near a zero norm, the projector definition itself becomes ill-conditioned and should trigger a diagnostic rather than a formal derivative.

11 Exact and practical forms of the band response

11.1 The induced Kohn-Sham potential

The exact response matrix in Eq. (26) contains

$$H_A = H_A^{\mathrm{Pulay}} + V_{\mathrm{KS},A}. \tag{67}$$

In the ordinary stationary force, the $V_{\mathrm{KS},A}$ contribution cancels against the density-functional variation. In the spectral-subspace response Q_A, however, H_A enters as a wavefunction-response operator. There is no general proof that the induced potential can be discarded from Eq. (26).

The practical first implementation can use the already available H_A^{Pulay} and O_A. This is a fixed-potential response approximation. Its accuracy must be measured, rather than assumed, by comparing the resulting U_A^{win} and forces against finite differences of the projector at self-consistency.

11.2 Stored-band sum versus Sternheimer equation

Formally the complement response can be written with a reduced resolvent,

$$|\psi_{n,A}\rangle_{\perp W} = Q_R(\epsilon_n - H)_R^{-1} Q_R(H_A - \epsilon_n O_A)|\psi_n\rangle, \tag{68}$$

which is the Sternheimer formulation. The explicit stored-band sum is the spectral representation of the same resolvent truncated to available eigenstates. Since LAPW1 already writes many bands, the initial implementation should use Eq. (26) directly and increase the number of LAPW1 bands until the correction plateaus.

Only if the plateau cannot be reached economically should an iterative Sternheimer solver be considered.

12 Mapping to the present `dmft2` implementation

Location or object	Present role	Required extension or correction
`Read_Vec_Spin` in `l2mainSubs.f90`	Reads and stores all eigenvectors available in `case.vector[so]`.	Reuse the stored included and excluded bands; no new eigensolver is initially required.
`Build_DMFT_Projector`	Builds projectors only for `DM_nemin:DM_nemax`; produces ordinary and vector channels.	Also build raw ordinary projectors U_{am} for stored excluded bands. Keep raw and normalized objects conceptually distinct.
`cmp_overlap`	Accumulates S over $\mathbf{k}$ and computes `SOlapm`$= S^{-1/2}$.	Accumulate S_A with the same weights, band indices, correlated-block filtering, and projector convention.

Location or object	Present role	Required extension or correction
`RenormalizeTrans`	Right-multiplies every `nipc` channel by X.	Keep for the fixed-X part. Add the separate UX_A derivative contribution; do not imitate it by scalar rescaling.
`CompressSigma-Transformation2`	Forms ordinary and vector `lgTrans` from already normalized channels.	The existing direct term remains usable after removal of the later diagonal division. New general derivative contractions should be matrix based.
`cmpLogGdloc`	Evaluates the logarithmic term and current orbital/projector force, presently in the diagonal-self-energy branch.	Set the post-factor to one; remove dependence on diagonal `SOlapm` elements. Add window and X_A contractions with the same Matsubara-tail treatment.
`Force1_DMFT`, `Force2`, `Force3`, surface terms	Contract LAPW bilinear kernels with ρ and $\rho\epsilon$ to make ordinary Pulay forces.	Expose or reconstruct the rectangular $H_A^{R,W}$ and $O_A^{R,W}$ blocks before final density-matrix contraction.
`Aweight`, `AEweight`, `wEpsw`	Carry full off-diagonal DMFT density and energy-weighted density matrices.	Confirm that the same band ordering and generalized-overlap convention are used in the response blocks.
`forb(3,natm)`	Current projector force is assigned only to correlated atoms.	Accumulate F_A^{win} and F_A^X for every inequivalent displaced atom.

The existing code already demonstrates that off-diagonal force information is needed: `Aweight`, `AEweight`, and `wEpsw` are full matrices, and the LAPW radial coefficients are rotated with these matrices. The missing engineering step is to retain the bilinear force kernels as the rectangular response blocks of Eq. (29), rather than only returning their final contraction with a density matrix.

13 Proposed implementation sequence

The implementation should be staged so that each new mathematical object can be tested independently.

Stage 0: remove the inconsistent post-factor

For the force path in `cmpLogGdloc`, impose

$$\texttt{renormalize} = 1. \tag{69}$$

Do not otherwise alter the current vector-projector force. Repeat existing force tests with and without `SOlapm.dat`. This isolates the effect of the erroneous double normalization.

Stage 1: expose all-band raw projectors

At every $\mathbf{k}$ build

$$U_W^{\mathbf{k}} \in \mathbb{C}^{N_W \times N_{\mathrm{loc}}}, \qquad U_R^{\mathbf{k}} \in \mathbb{C}^{N_R \times N_{\mathrm{loc}}} \tag{70}$$

from bands already present in `case.vector[so]`. Verify that restricting the all-band object back to W reproduces the old projector bit for bit before normalization.

Stage 2: construct rectangular LAPW response blocks

For each atom and Cartesian direction, construct

$$H_A^{R,W}, \qquad O_A^{R,W}, \qquad M_A^{R,W} = H_A^{R,W} - O_A^{R,W}\,\mathrm{diag}(\epsilon_W). \tag{71}$$

Use the same radial, nonspherical, interstitial, and surface conventions as the ordinary force. Initially omit $V_{\mathrm{KS},A}$ but label the result explicitly as fixed-potential response.

Stage 3: compute the finite-window projector derivative

Evaluate

$$C_{an}^A = \frac{M_{an}^A}{\epsilon_n - \epsilon_a}, \qquad U_A^{\mathrm{win}} = C_A^\dagger U_R. \tag{72}$$

Here the second equation is the matrix form of Eq. (32). Store or immediately contract U_A^{win} atom by atom to control memory use.

Stage 4: accumulate S_A and solve for X_A

For every displaced atom/component accumulate

$$S_A = \sum_{\mathbf{k}} w_{\mathbf{k}} (U_A^\dagger U + U^\dagger U_A), \tag{73}$$

perform the MPI reduction, and solve Eq. (44). Check Hermiticity and Eq. (50) before computing any force.

Stage 5: add F_A^{win} and F_A^X

Use Eq. (56) for the window term. Accumulate K from Eq. (59) during the frequency/$\mathbf{k}$ loop, then evaluate Eq. (60). The result should be stored as a separate diagnostic force component during development:

$$F_A^{\mathrm{proj}} = F_A^{\mathrm{dir}} + F_A^{\mathrm{win}} + F_A^X \quad (+F_A^{dff} \text{ when applicable}). \tag{74}$$

Stage 6: add optional $d_{n,A}$ and full response refinements

If projector types 2 or 4 are used, implement Eq. (66). Then assess the need for $V_{\mathrm{KS},A}$ by finite differences. Do not combine these refinements with the first implementation before the window and X_A pieces are independently validated.

14 Memory and computational cost

Let N_W be the DMFT-window size, N_R the number of stored excluded bands, N_L the local correlated dimension, and N_A the number of inequivalent atom-direction perturbations.

The dominant new algebra per $\mathbf{k}$ and perturbation is approximately

$$O(N_R N_W N_L) \tag{75}$$

for forming $U_A^{\mathrm{win}} = C_A^\dagger U_R$, after the rectangular response matrix is available. The Sylvester solve costs $O(N_L^3)$ and is negligible.

It is not necessary to store C_A for all atoms simultaneously. One can loop over atoms/components, construct the rectangular block, accumulate S_A and F_A^{win}, then release the block. Likewise, K is only an $N_L \times N_L$ matrix for each correlated block.

This explicit-band approach is expected to be much cheaper than a separate iterative Sternheimer solve for every occupied/window state and every atom-direction perturbation.

15 Verification strategy

15.1 Algebraic checks at each geometry

Require

$$S^\dagger = S, \tag{76}$$
$$S_A^\dagger = S_A, \tag{77}$$
$$X^\dagger = X, \tag{78}$$
$$X_A^\dagger = X_A, \tag{79}$$

and monitor the residuals

$$r_X = \|Y X_A + X_A Y + X S_A X\|, \tag{80}$$
$$r_N = \left\| \sum_{\mathbf{k}} w_{\mathbf{k}} (D_A^{\mathbf{k}\dagger} \overline{U}^{\mathbf{k}} + \overline{U}^{\mathbf{k}\dagger} D_A^{\mathbf{k}}) \right\|. \tag{81}$$

Both should be near machine precision relative to the corresponding matrix norms, apart from deliberate truncation in the band response.

15.2 Finite-difference checks of intermediate objects

For a small displacement δ and a fully reconverged electronic state, compare

$$U_A^{\mathrm{FD}} = \frac{U(R_A+\delta) - U(R_A-\delta)}{2\delta}, \tag{82}$$

$$S_A^{\mathrm{FD}} = \frac{S(R_A+\delta) - S(R_A-\delta)}{2\delta}, \tag{83}$$

$$X_A^{\mathrm{FD}} = \frac{X(R_A+\delta) - X(R_A-\delta)}{2\delta}. \tag{84}$$

Because individual eigenvectors have arbitrary phases and can rotate within degenerate subspaces, compare gauge-invariant quantities first: Q_A, S_A, X_A, and the final force. If comparing U_A directly, align the perturbed window states by a maximum-overlap/parallel-transport rotation.

15.3 Band-completeness test

Increase the highest stored LAPW1/LAPWSO band and monitor

$$F_A^{\mathrm{win}}(N_R), \qquad S_A^{\mathrm{win}}(N_R), \qquad \sum_A \mathbf{F}_A(N_R). \tag{85}$$

The results should approach a plateau. A nonsmooth projector at the muffin-tin boundary may require many bands, so convergence should be judged empirically.

In the formal complete-band limit,

$$Q_{\mathbf{k}} \to I \quad \Longrightarrow \quad Q_{A,\mathbf{k}} \to 0, \qquad U_A^{\mathrm{win}} \to 0. \tag{86}$$

This provides a conceptual limit, although the practical DMFT window is not usually taken to the complete Hilbert space.

15.4 Acoustic sum rule

Under a rigid translation of all nuclei and all local orbitals, the normalized projector geometry is unchanged. Therefore

$$\boxed{\sum_A S_A = 0, \qquad \sum_A X_A = 0, \qquad \sum_A F_A^{\mathrm{proj}} = 0,} \tag{87}$$

up to basis, band, $\mathbf{k}$, and self-consistency errors. The acoustic sum is not a substitute for the atom-resolved finite-difference test, but it is an excellent global diagnostic for a term assigned only to correlated atoms when it should act on all atoms.

15.5 Recommended sequence of physical tests

1. One correlated atom per cell, where the old implementation already behaves well.

2. All atoms correlated, to test the normalization algebra without a correlated/uncorrelated partition.
3. Two dissimilar atoms with only one correlated, the configuration that exposes the present force-sum problem.
4. Repeat case 3 while increasing the stored-band complement.
5. Repeat with $X = I$ only as a diagnostic, not as the final projector definition.

For every test print the decomposition

$$\mathbf{F}_A^{\text{total}} = \mathbf{F}_A^{\text{ordinary Pulay}} + \mathbf{F}_A^{\text{dir}} + \mathbf{F}_A^{\text{win}} + \mathbf{F}_A^{X} + \mathbf{F}_A^{\text{HF}} + \cdots \tag{88}$$

so that cancellations and missing atom assignments remain visible.

16 Controlled approximations and open questions

1. **Induced potential.** The exact C_{an}^{A} contains $V_{\text{KS},A}$. The fixed-potential response is a practical first approximation, not a formal identity.
2. **Finite stored complement.** The explicit sum is exact only in a complete complement. Convergence with the LAPW1 band count is mandatory.
3. **Near-degenerate window boundary.** Small denominators require an enlarged window or block treatment.
4. **Band-dependent `dff`.** Its derivative matters for projector types 2 and 4 and should be tested separately.
5. **General self-energy matrices.** The current logarithmic force is evaluated in a branch that assumes a diagonal self-energy index structure. Equations (14), (56), and (57) are matrix general and should be the long-term implementation target.
6. **Consistency of `SOlapm.dat`.** A saved X must be built from exactly the same band window, $\mathbf{k}$ weights, projector type, spin convention, and correlated-block filtering used to construct S_A.

17 Compact formula sheet

For implementation, the complete chain is:

$$U_{nm}^{\mathbf{k}} = \langle \psi_{n\mathbf{k}} | \phi_m \rangle, \tag{89}$$

$$S = \sum_{\mathbf{k}} w_{\mathbf{k}} U^{\mathbf{k}\dagger} U^{\mathbf{k}}, \tag{90}$$

$$X = S^{-1/2}, \qquad \overline{U}^{\mathbf{k}} = U^{\mathbf{k}} X, \tag{91}$$

$$C_{an}^{A,\mathbf{k}} = \frac{A_{a\mathbf{k}}^{\dagger}(H_{A,\mathbf{k}} - \epsilon_{n\mathbf{k}} O_{A,\mathbf{k}}) A_{n\mathbf{k}}}{\epsilon_{n\mathbf{k}} - \epsilon_{a\mathbf{k}}}, \tag{92}$$

$$(U_A^{\mathbf{k},\text{win}})_{nm} = \sum_{a \in R_{\mathbf{k}}} C_{an}^{A,\mathbf{k}*} U_{am}^{\mathbf{k}}, \tag{93}$$

$$U_A^{\mathbf{k}} = U_A^{\mathbf{k},\mathrm{dir}} + U_A^{\mathbf{k},\mathrm{win}}, \tag{94}$$

$$S_A = \sum_{\mathbf{k}} w_{\mathbf{k}} (U_A^{\mathbf{k}\dagger} U^{\mathbf{k}} + U^{\mathbf{k}\dagger} U_A^{\mathbf{k}}), \tag{95}$$

$$Y X_A + X_A Y = -X S_A X, \qquad Y = S^{1/2}, \tag{96}$$

$$D_A^{\mathbf{k}} = U_A^{\mathbf{k}} X + U^{\mathbf{k}} X_A, \tag{97}$$

$$F_A^{\mathrm{proj}} = -T \sum_{\mathbf{k}, i\omega_n} w_{\mathbf{k}} \,\mathrm{Tr}_b \Big[G_{\mathbf{k}} (D_A^{\mathbf{k}} \widetilde{\Sigma} \overline{U}^{\mathbf{k}\dagger} + \overline{U}^{\mathbf{k}} \widetilde{\Sigma} D_A^{\mathbf{k}\dagger}) \Big]. \tag{98}$$

The correction relative to the current fixed-X vector-projector force is

$$\boxed{\Delta F_A^{\mathrm{proj}} = F_A^{\mathrm{win}} + F_A^{X} \quad (+F_A^{dff} \text{ for projector types } 2/4),} \tag{99}$$

and the later elementwise logarithmic-force rescaling must be

$$\boxed{\texttt{renormalize} = 1.} \tag{100}$$

18 References and implementation sources

1. K. Haule, *Some notes on forces in LAPW*, especially the stationary-functional variation, Eqs. 326–365, and the implementation of the explicit DMFT projector term.
2. `/Users/haule/projects/eDMFT_new/src/dmft2/l2mainSubs.f90`: vector-file reading, projector construction, overlap accumulation, $S^{-1/2}$, normalization, and compressed transformations.
3. `/Users/haule/projects/eDMFT_new/src/dmft2/cmp_log_gdloc.f90`: Matsubara logarithm and present projector-force contraction.
4. `/Users/haule/projects/eDMFT_new/src/dmft2/forces.f90` and `alm_blm_clm.f90`: ordinary off-diagonal DMFT Pulay-force kernels.

Status: derivation and implementation plan only. No source-code changes have been made as part of this document.